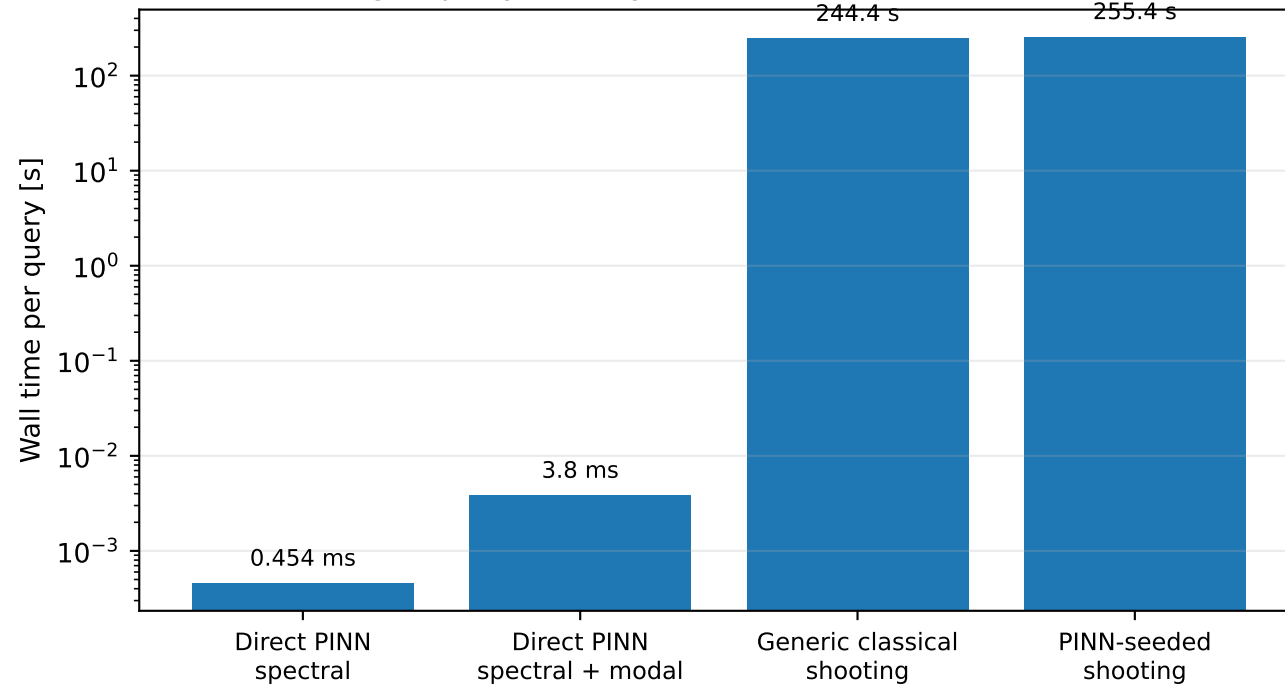
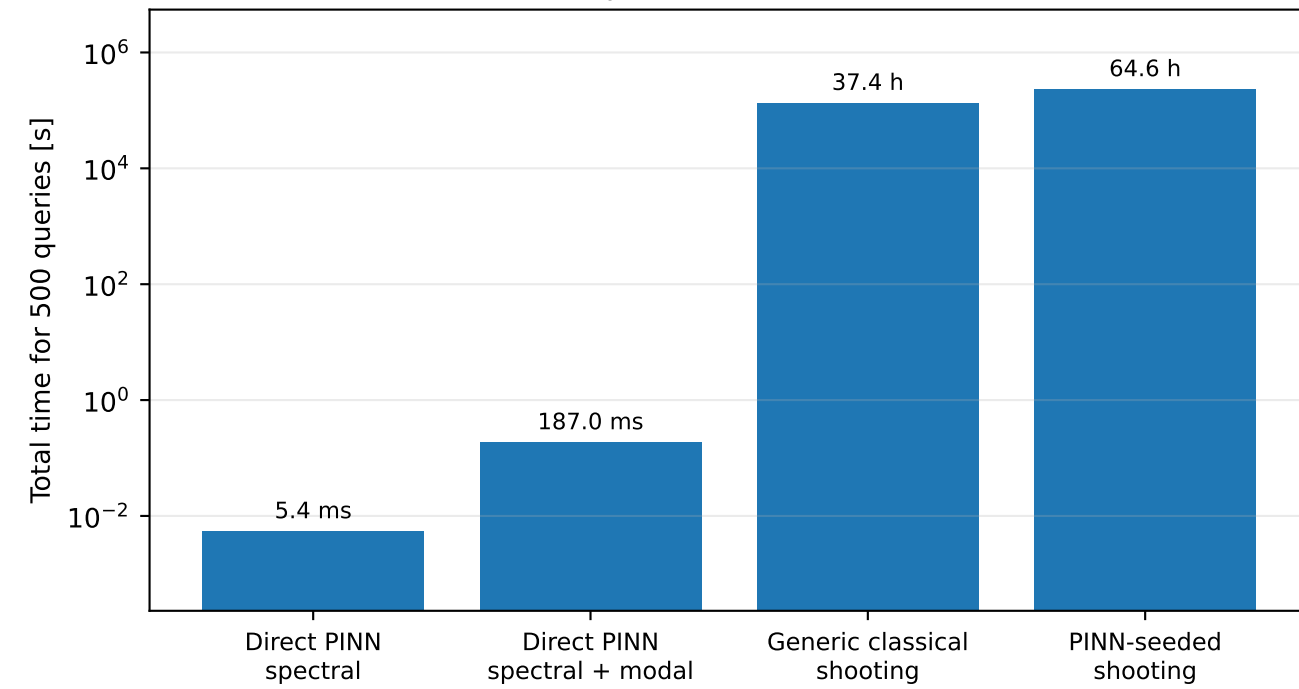


Computational cost and robustness of the supersonic spectral pipelines

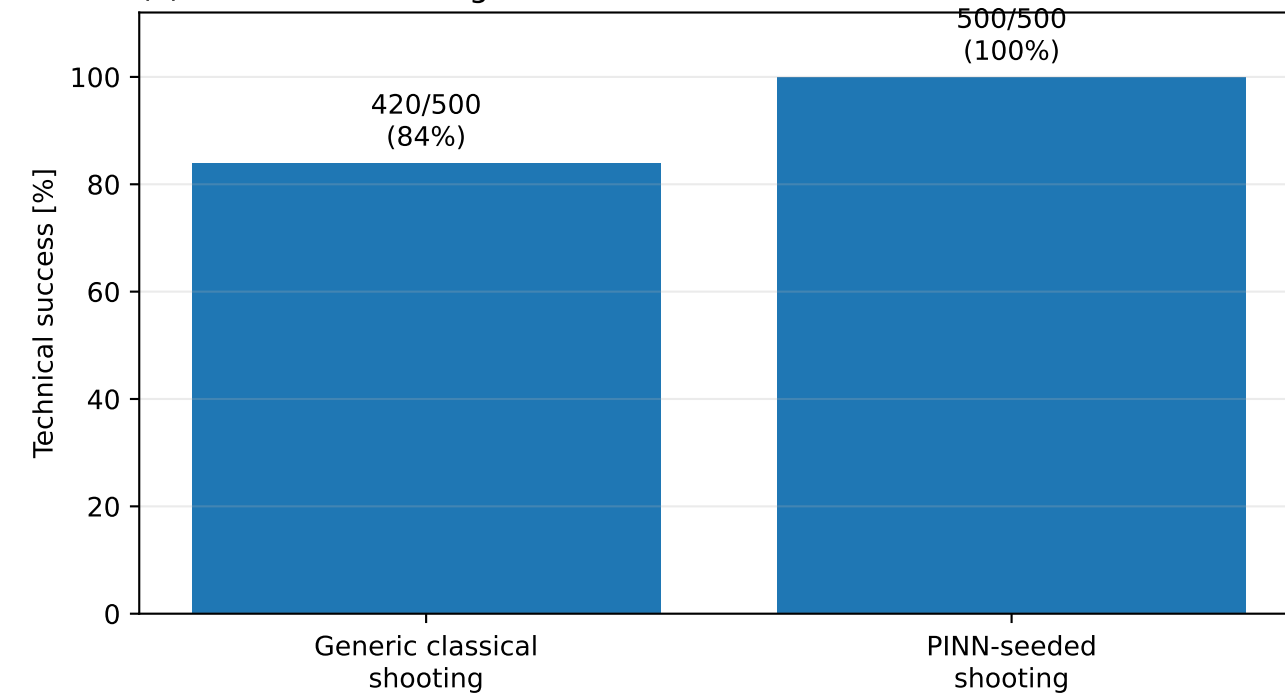
(a) COST1: single-query latency



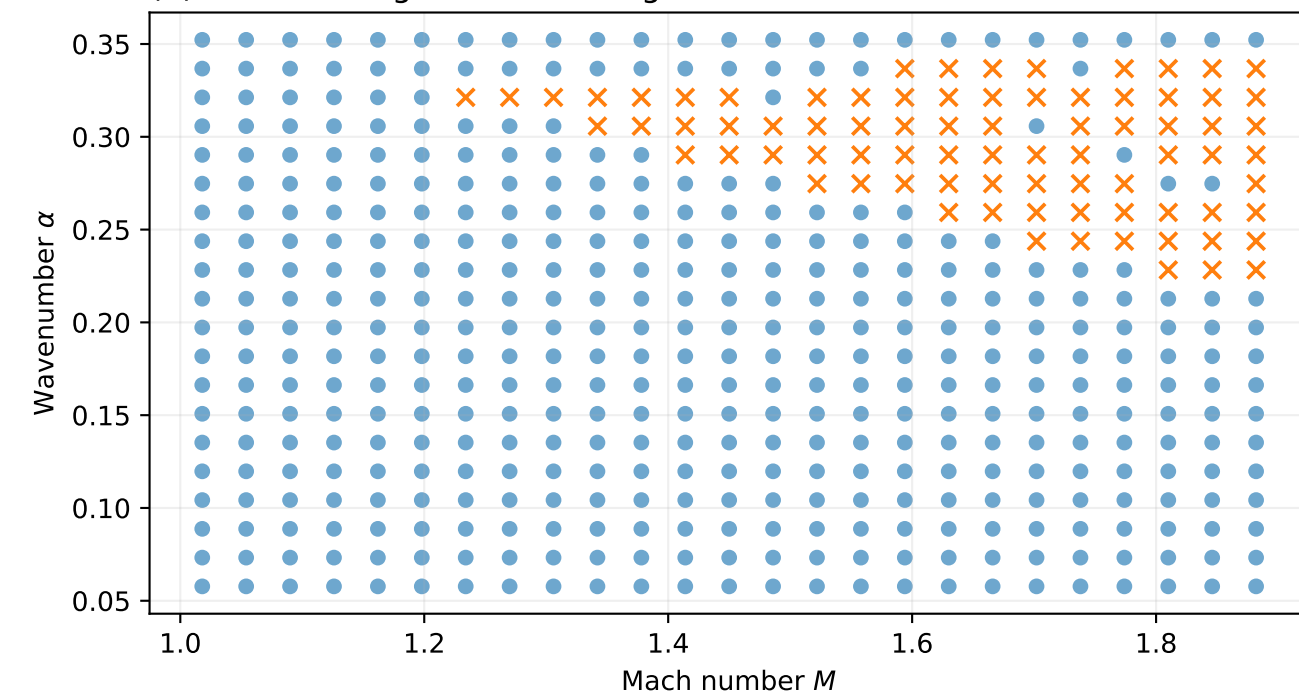
(b) COST500: batch/serial-equivalent cost



(c) COST500 shooting robustness



(d) Location of generic-shooting failures



● Generic shooting: success ✕ Generic shooting: failure

Timing convention: direct PINN = GPU-batched wall time; shooting methods = serial-equivalent sum of individual solves.